

Reflection, Analysis and Interpretation

1. What was the most challenging part?

The most challenging part of this project was understanding the theoretical relationship between Shannon Information, entropy, cross Entropy, softmax and neural network predictions. At first these concepts seemed like separate mathematical formulas but during the project I learned that they are closely connected.

Another challenge was implementing entropy and cross entropy manually instead of using built in library functions. This required understanding why numerical issues such as $\log(0)$ which is comes to negative infinity occur and how adding a small epsilon value prevents computation errors.

2. What did I learn about information theory and ML?

This project helped me understand several important concepts from Information Theory and Machine Learning.

First I learned that information measures surprise which is Events that are less likely contain more information because they are more unexpected.

Second I learned that entropy measures uncertainty in a probability distribution. High entropy means the probabilities are spread across many classes and the model is uncertain. Low entropy means the model strongly favors one class and is confident.

Third I learned that cross entropy measures how well the predicted probability distribution matches the true distribution. During training the model minimizes cross entropy which encourages it to assign higher probability to the correct class.

I also learned the role of Softmax. Neural networks produce logits which are not valid probabilities. Softmax converts these logits into probabilities that sum to one allowing entropy and cross entropy to be calculated.

3.How did my understanding change after doing this task?

Before completing this project I viewed loss functions mainly as mathematical formulas used during training. I understood how to use them but not why they worked.

After completing the project I now understand that cross entropy directly measures the amount of surprise associated with the true class according to the model's predictions. A lower cross entropy means the model assigns higher probability to the correct class.

I also gained a deeper understanding of model confidence. Before this task I assumed that confident predictions were usually correct. Through the entropy analysis I discovered that a model can be confidently wrong. This showed me that confidence and correctness are not the same thing.

4. Analysis and Interpretation of my work.

The trained MLP achieved a test accuracy of 86.72% on the Fashion MNIST dataset which shows that it learned meaningful patterns from the images.

The average entropy for correct predictions was 0.228, while the average entropy for incorrect predictions was 0.785. This suggests that the model was generally more uncertain when making incorrect predictions. Therefore, entropy was useful as a measure of prediction confidence.

The average cross entropy for correct predictions was 0.096, while the average cross entropy for incorrect predictions was 2.210. The large difference shows that cross entropy strongly penalizes predictions that assign low probability to the true class.

The histogram of entropy values showed that most predictions had relatively low entropy, meaning the model was often confident. The boxplots further confirmed that incorrect predictions generally had higher entropy and higher cross entropy values than correct predictions.

One particularly interesting result was the presence of confidently wrong predictions. Some samples had extremely low entropy but were still classified incorrectly. For example one sample had an entropy of only 0.0015 while producing a cross entropy loss greater than 10. This demonstrates that low entropy indicates confidence but it does not guarantee correctness.

Overall, the experiment confirmed that entropy is useful for measuring uncertainty, while cross entropy is useful for measuring prediction quality and guiding learning during training.

In Conclusion

I felt excited in understanding how these formulas come about and how each of them are connected and build on one another. In previous use of training a model I just used the loss function just because it was the recommended one not because I know why but after this training I understand that there is a lot to learn in how different methods come about whether for loss function, activation function why use softmax and different ML concepts.